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MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : M-402

MATHEMATICS - III

Time Allotted : 3 Hours | *Full Marks* : 70

The figures in the margin indicate full marks.

GROUP – A

(Objective Type Questions)

1. Answer any *ten* of the following : $10 \times 2 = 20$

i) If $f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi \end{cases}$ be represented in Fourier's series $\frac{1}{2}a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$, then find a_0.

ii) If $F(s)$ represents the Fourier transform of the function $f(t)$, find the Fourier transform of $f(t) \cos \omega t$, ω being a constant.

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iii) Find the value of the integral $\int_C \frac{dz}{z+3}$, where the circle $C : |z| = 1$ is traversed in the counter clockwise sense.

iv) Find the value of m, so that the function $3y - 5x^2 + my^2$ is a harmonic function.

v) For any event $A \subseteq S$ (certain event), prove that $P(\bar{A}) = 1 - P(A)$.

vi) Find the nature and location of singularities of the function $\frac{e^{2z}}{(z-1)^3}$.

vii) Find the residue of $f(z) = \frac{z^2}{(z+1)^2(z-2)}$ at $z = -1$.

viii) Find $P_n(1)$.

ix) Show that two independent events with non-zero probabilities cannot be mutually exclusive.

x) A determinant is chosen at random from the set of all determinants of order 2 with elements 0 or 1 only. Find the probability that the value of the determinant chosen is positive.

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xii) If $f(x) = kx$, $x = 1, 2, \ldots, n$ be the *p.m.f.* of a random variable X, then find the mean of X.

xiii) Probability density function of a random variable X is given below :

$$f(x) = \begin{cases} 0 \cdot 25, & 1 \le x \le 5 \\ 0, & \textit{otherwise} \end{cases}$$. Find $P(X \le 4)$.

xiii) Determine the singular points of the differential equation $x(x-2)^3 \frac{d^2y}{dx^2} + 3(x-2)^3 \frac{dy}{dx} + 4y = 0$ and specify whether they are regular or irregular.

xiv) Express $J_2(x)$ in terms of $J_0(x)$ and $J_1(x)$.

GROUP – B

Answer any *five* questions taking at least *one* question from each Module : $5 \times 10 = 50$

Module - I

2. a) Find the Fourier series of $f(x) = x$ in the interval $-\pi \le x \le \pi$ and hence deduce that

$$1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots \text{ to } \infty = \frac{\pi}{4}.$$

b) Find the Fourier cosine transform of $f(x) = e^{-x}$, $x \ge 0$. 6 + 4

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3. a) Obtain the Fourier *sine* transform of

$$f(x) = \begin{cases} 1, & 0 < x < 1 \\ 0, & \textit{otherwise} \end{cases}$$.

b) Prove that the Heaviside's unit step function $H(t)$ does not possess Fourier transform. 6 + 4

4. a) Using Parseval's identity, prove that

$$\int_0^\infty \frac{dx}{(x^2+1)^2} = \int_0^\infty \frac{x^2}{(x^2+1)^2}\, dx = \frac{\pi}{4}.$$

b) Find the inverse Fourier transform of $\frac{1}{s^2 + 4s + 13}$.

6 + 4

Module - II

5. a) The real part of an analytic function $f(z)$ where $z = x + iy$ is given by $e^{-y} \cos x$. Find the imaginary part of $f(z)$.

b) Evaluate $\int_0^{3+i} (\bar{z})^2\, dz$ along the line $3y = x$.

5 + 5

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6. a) Prove that $\oint_C \frac{z+1}{z^2-2z}\,dz = 2\pi i$ where C is the circle $|z| = 5$.

b) Expand $f(z) = \frac{1}{(z+1)(z+3)}$ in Laurent's series valid for the region $0 < |z+1| < 2$. 5 + 5

7. a) If $f(z)$ is an analytic function of z then show that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |Re f(z)|^2 = 2|f'(z)|^2$. 5

b) Prove that $u(x, y) = e^{-x}(x \sin y - y \cos y)$ is harmonic function and find its conjugate harmonic. 2 + 3

Module - III

8. a) The probability that a student knows the correct answer to a multiple choice question is 2/3. If the student does not know the answer, then the student guesses the answer. The probability of the guessed answer being correct is 1/4. Given that the student has answered the question correctly, find the conditional probability that the student knows the correct answer.

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b) A random variable X has probability density function $f(x)$ as given below :

$$f(x) = \begin{cases} a + bx, & 0 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

If the expected value of $E[X] = \frac{2}{3}$, then find

$Pr[X < 0{\cdot}5)$. 5 + 5

9. a) A box contains 10 pair of shoes. If 8 shoes are randomly selected, what is the probability that there is no complete pair and exactly one complete pair.

b) If the random variable X takes the values 1, 2, 3, and 4 such that

$2P(X = 1) = 3P(X = 2) = P(X = 3) = 5P(X = 4)$,

find the probability distribution of X. 5 + 5

10. a) In a Normal distribution 31% items are under 45 and 8% are above 64. Find the mean and standard deviation.

[Given : $P(0 < z < 1{\cdot}405) = 0{\cdot}42$,

$P(-0{\cdot}496 < z < 0) = 0{\cdot}19$].

b) If the random variable X follows Binomial distribution with parameters n and p then prove that mean is np and variance is $np(1-p)$. 5 + 5

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Module - IV

11. a) Using Fourier *sine* transform method, solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$, $(x > 0, t > 0)$ subject to the conditions $u(0, t) = 0$,

$u(x, 0) = \begin{cases} 1, & 0 < x < 1 \\ 0, & x \geq 1 \end{cases}$ & $u(x, t)$ is bounded.

b) Solve by using the method of separation of variables $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u$, where $u(x, 0) = 6e^{-3x}$.

6 + 4

12. a) Obtain the series solution of the differential equation $\frac{d^2y}{dx^2} + y = 0$ near $x = 0$ such that

$y(0) = 1$, $y'(0) = 2$.

b) Prove that : $\int_{-1}^{1} P_m(x) . P_n(x) dx = 0$ if $m \neq n$.

6 + 4

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